

Polar and Exponential Forms of Complex Numbers

Rail Engineering Mathematics

Sheffield Hallam University

Lecture 5

Learning Outcomes

- Represent complex numbers in an Argand diagram.
- Express complex numbers in Cartesian (rectangular), polar and exponential form and convert between them.

Note: Make sure your calculator is set to **radians**, not degrees!

Argand Diagrams

Complex numbers written in the form

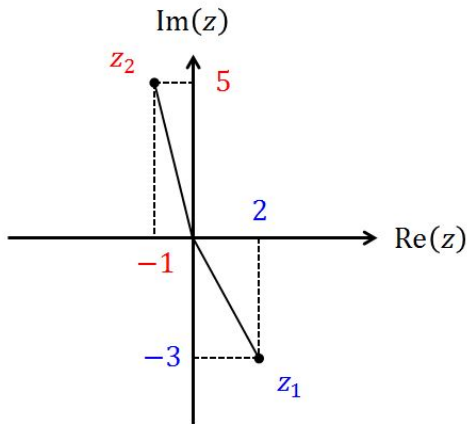
$$z = x + jy$$

are said to be in **Cartesian** form (also called rectangular form).

In this form we can represent a complex number graphically using the co-ordinate (x, y) in an Argand diagram, where the x -axis is the *real* part and the y -axis represents the *imaginary* part.

Argand Diagrams

Plotting $z_1 = 2 - j3$ and $z_2 = -1 + j5$ in an Argand diagram:



Modulus and Argument

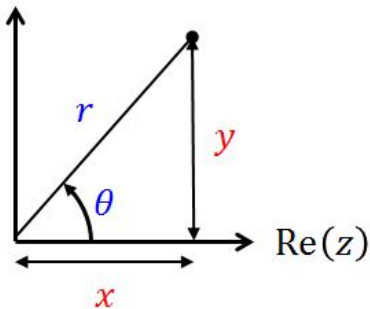
The Argand diagram suggests an alternative way of representing complex numbers.

Instead of using the co-ordinates (x, y) to fix the position of the end of a line in the Argand diagram, we could define the line's position using the **modulus** r (length of the line from the origin) and the **argument** θ (angle relative to the positive real axis).

Polar Form

Given a complex number, $z = x + jy$, where both $x, y > 0$:

Im(z)



We can calculate the modulus using Pythagoras:

$$r = \sqrt{x^2 + y^2}$$

and the argument can be calculated using trigonometry:

$$\theta = \tan^{-1} \left(\frac{y}{x} \right)$$

Polar Form

So, if $z = x + jy$ is a complex number written in Cartesian form, then

Polar form:

$$\begin{aligned} z &= r \cos \theta + jr \sin \theta \\ &= r(\cos \theta + j \sin \theta) \end{aligned}$$

where r is the modulus and θ is the argument.

Polar Form

The shorthand form for $z = r(\cos \theta + j \sin \theta)$ is

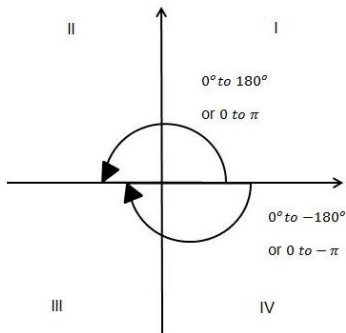
$$z = r \angle \theta \quad \text{when dealing with complex numbers.}$$

Or

$$P = (r, \theta) \quad \text{when specifying a point in polar co-ordinates.}$$

Measuring the Argument

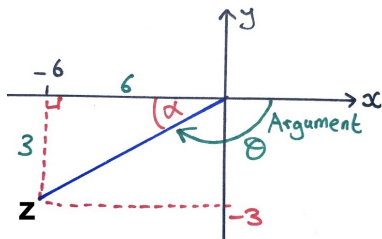
If z is not in the first or fourth quadrant, we need to do more to find the argument, as it is measured **anti-clockwise** from the **positive real axis**. By convention, it should be in the range $-\pi < \theta \leq +\pi$:



Measuring the Argument - an example:

Consider:

$$z = -6 - j3$$



Using right angle trig. we initially determine the angle α by:

$$\begin{aligned}\alpha &= \tan^{-1} \left(\frac{y}{x} \right) = \tan^{-1} \left(\frac{-3}{-6} \right) \\ &= 0.464\end{aligned}$$

But this is not the argument, rather:

$$\theta = \alpha - \pi = -2.678$$

This complex number is in the third quadrant.

Measuring the Argument

The general approach is to *always* calculate θ according to:

$$\theta = \tan^{-1} \left(\frac{y}{x} \right)$$

and then:

- If the complex number lies in **quadrant II** on the Argand diagram, then we **add 180° or π** to the result.
- If the complex number lies in **quadrant III** on the Argand diagram, then **subtract 180° or π** from the result.

Example 1

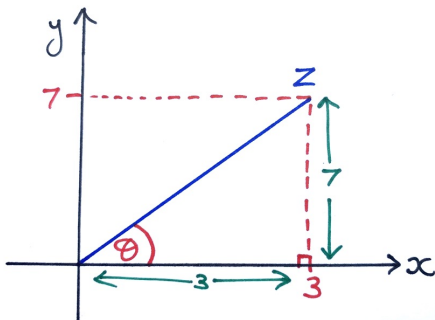
Express the following complex numbers in polar form:

1) $z = 3 + j7$

2) $z = -4 + j3$

Example 1 - Solutions

1)



Modulus:

$$\begin{aligned} r &= \sqrt{3^2 + 7^2} \\ &= \sqrt{58} \\ &= 7.616 \end{aligned}$$

Argument:

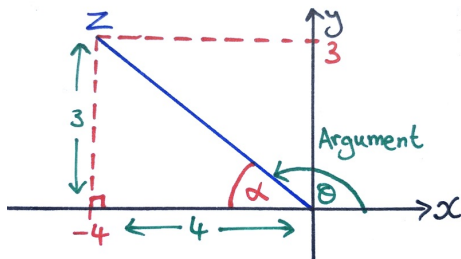
$$\theta = \tan^{-1} \left(\frac{7}{3} \right) = 1.166$$

Hence,

$$z = 7.616 \angle 1.166$$

Example 1 - Solutions

2)



Modulus:

$$\begin{aligned} r &= \sqrt{(-4)^2 + 3^2} \\ &= \sqrt{25} \\ &= 5 \end{aligned}$$

Argument:

$$\alpha = \tan^{-1} \left(\frac{3}{-4} \right) = -0.644$$

$$\theta = -0.644 + \pi = 2.498$$

Hence, $z = 5 \angle 2.498$

Converting to Cartesian Form

Given a complex number written in polar form:

$$z = r(\cos \theta + j \sin \theta)$$

We are easily able to convert to Cartesian (rectangular) form, using trigonometry, with the formulae:

$$x = r \cos \theta \quad \text{and} \quad y = r \sin \theta$$

Note that in this case in **does not matter which quadrant** the complex number lies in.

Example 2

Express the following complex numbers in Cartesian form:

1) $z = 8\angle 2.1$

2) $z = 5.3\angle -3$

Example 2 - Solutions

1)

$$x = 8 \cos(2.1) = -4.039 \quad \text{and} \quad y = 8 \sin(2.1) = 6.906$$

$$\therefore z = -4.039 + j6.906$$

2)

$$x = 5.3 \cos(-3) = -5.247 \quad \text{and} \quad y = 5.3 \sin(-3) = -0.748$$

$$\therefore z = -5.247 - j0.748$$

Exponential Form

It can be shown using Maclaurin series' (not a topic we'll encounter) that

Exponential Form:

$$r e^{j\theta} = r \cos(\theta) + j r \sin(\theta)$$

where r is the modulus and θ is the argument.

But the RHS of this is a polar form complex number, hence if

$$z = x + jy \text{ and } |z| = r \text{ while } \arg(z) = \theta,$$

then $z = r \cos(\theta) + j r \sin(\theta)$ (polar form) and $z = r e^{j\theta}$ (exponential form).

Example 3

Rewrite $z = 5\angle\frac{\pi}{6}$ in exponential form.

Example 3 - Solution

In expanded polar form, this can be written as

$$z = 5 \cos \left(\frac{\pi}{6} \right) + j5 \sin \left(\frac{\pi}{6} \right).$$

Here $r = 5$ and $\theta = \frac{\pi}{6}$.

Therefore, this can be expressed in exponential form as

$$z = 5 e^{j \frac{\pi}{6}}$$

Polar/Exponential Form Arithmetic

Polar/exponential form can be useful since multiplications and division in polar form are much easier; as shown by the formulae:

$$\begin{aligned} z_1 z_2 &= r_1 r_2 \angle (\theta_1 + \theta_2) \quad \text{or} \\ &= r_1 r_2 e^{j(\theta_1 + \theta_2)} \end{aligned}$$

and

$$\begin{aligned} \frac{z_1}{z_2} &= \frac{r_1}{r_2} \angle (\theta_1 - \theta_2) \quad \text{or} \\ &= \frac{r_1}{r_2} e^{j(\theta_1 - \theta_2)} \end{aligned}$$

where $z_1 = r_1 \angle \theta_1$ and $z_2 = r_2 \angle \theta_2$.

Example 4

Given that $z_1 = 5.3 \angle 2.1$ and $z_2 = 2.7 \angle -0.3$, determine:

1) $z_1 z_2$

2) $\frac{z_1}{z_2}$

Example 4 - Solutions

1)

$$\begin{aligned} z_1 z_2 &= (5.3 \times 2.7) \angle (2.1 + (-0.3)) \\ &= 14.31 \angle 1.8 \end{aligned}$$

2)

$$\begin{aligned} \frac{z_1}{z_2} &= \left(\frac{5.3}{2.7} \right) \angle (2.1 - (-0.3)) \\ &= 1.963 \angle 2.4 \end{aligned}$$